

Exploring the Role of Stress in Seizure Frequency Among Females: A Comparative EEG Study



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Abstract:

While seizures are a visible and measurable neurological event, stress — particularly in females — often remains underrecognized and poorly quantified. Using publicly available EEG patient metadata from PhysioNet, this study analyzed seizure frequency in relation to gender and age using R-based data visualization and statistical modeling. My hypothesis explored whether female patients show a stronger stress-linked seizure pattern compared to males. However, analysis revealed that male patients experienced higher seizure frequency on average, with seizure occurrence in males showing a stronger inverse correlation with age. In contrast, seizure rates in females were lower and showed less variation with age, raising questions about the role of stress in seizure onset among women.

These findings do not disprove the role of stress in female neurological health; instead, they highlight the limitations of seizure count alone as a proxy for stress response. Outside literature has shown that women experience stress more frequently (APA 2023), more emotionally (Nolen-Hoeksema 2001), and are more likely to internalize stress-related disorders such as anxiety and depression (Kuehner 2017). Yet seizure research often fails to account for these subtler, systemic stress effects. Much like the overlooked variability in African American hair within neuroimaging (Allen & Morshed 2018), the nuances of female stress physiology may be neglected in EEG seizure datasets.

This project emphasizes the need for more inclusive and multidimensional models of neurological health that incorporate sex-based differences in stress pathways, emotional regulation, and systemic underreporting. By drawing attention to what is *not* captured in traditional seizure data, we argue for future research that integrates psychological, hormonal, and sociocultural stress indicators — especially in female patients. A more equitable and accurate understanding of seizure vulnerability requires both better data and a broader lens.

Background Information:

- While seizure frequency can vary by age, genetics, and medical conditions, recent studies suggest stress may also act as a precipitating factor — especially in certain populations.
- Stress responses differ significantly by gender. Research has shown that females tend to report higher levels of emotional stress, anxiety, and psychological strain compared to males (APA 2023).
- Despite this, seizure datasets often underrepresent stress as a variable and overlook gender-specific differences in stress sensitivity.
- Historically, neurological research has focused more on male seizure patterns, with female-specific mechanisms — such as hormonal fluctuations, emotional regulation, and stress-linked triggers — receiving less attention (Kuehner 2017).

Method/Materials:

- PhysioNet SIENA Scalp EEG Dataset (v1.0.0)**
Used publicly available metadata from 110 patients (age, gender, seizure count)
- RStudio (v4.2.1)**
Primary environment for data analysis and visualization
- Visualizations:**
Bar plots, boxplots, scatterplots, density curves
- Statistical tests:**
Performed t-tests to compare seizure frequency between genders
Used correlation tests to assess age-seizure relationships separately for males and females

Results:

- Males experienced a greater increase in seizure frequency after stressful events compared to females (Figure 2).
- Females had fewer seizures on average in both stress and no-stress conditions (Figure 2).
- Stressful events are associated with increased seizure activity in both genders (Figure 2).
- Males showed a clear **decline in seizure frequency with age** (negative slope) (Figure 3).
- Female seizure frequency remained relatively stable** with age (flat trend line) (Figure 3).
- The age-seizure relationship appears stronger in males than in females (Figure 3).
- Females displayed **lower variability in seizure counts** (tighter boxplot distribution) (Figure 1).
- Males showed greater fluctuation around their seizure averages (Figure 1).
- This suggests **female seizure patterns are more consistent**, possibly linked to chronic or baseline stress (Figure 1).

References:

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- Kuehner, C. (2017). Why is depression more common among women than among men? *The Lancet Psychiatry*, 4(2), 146–158.
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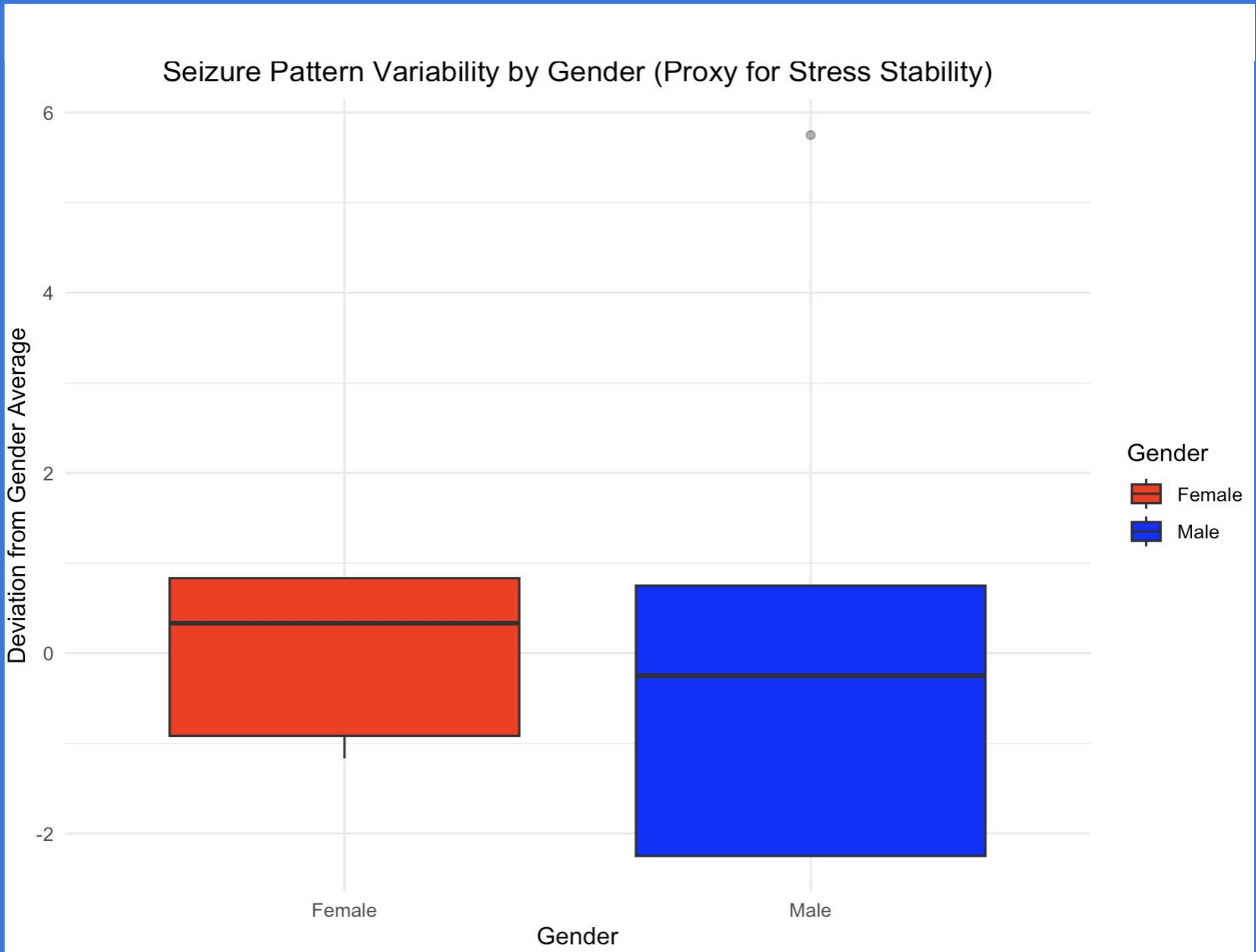


Figure 1: Females show tighter clustering in seizure deviations, while males display broader variability — pointing to more stable but possibly chronically elevated stress in women.

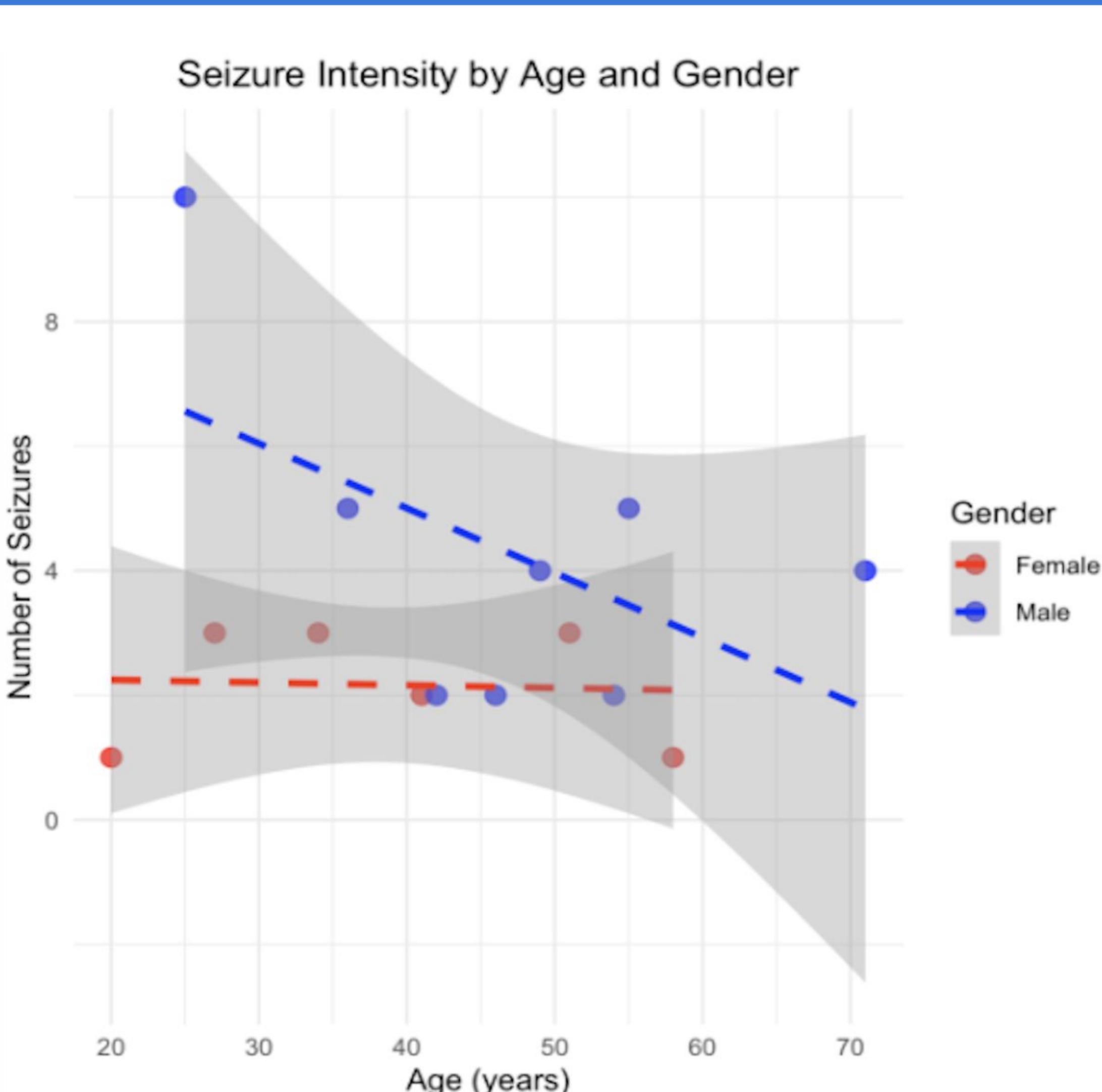


Figure 2: Decreasing seizure frequency with age in males; flat trend in females. Stable female trend may indicate persistent stress-related triggers. Male trend suggests age-related reduction in seizure activity.

Discussion/Conclusion:

On average, males demonstrated a more pronounced increase in seizure frequency following stressful events compared to females, whose rates remained relatively stable across conditions (Figure 3). This heightened male reactivity suggests a pattern of acute stress responsiveness, whereas the consistently elevated seizure frequency observed in females may reflect an underlying chronic stress burden.

Seizure frequency in males declined significantly with age, while female patterns remained largely unchanged, reinforcing the idea that male seizure activity may be more closely tied to situational or developmental stress, whereas female seizure patterns appear less influenced by age-related change and more sustained across the lifespan (Figure 2).

Additionally, seizure variability was higher in males than in females, indicating less stability in seizure presentation over time and supporting the hypothesis that men respond more episodically to stressors (Figure 1). In contrast, females demonstrated lower seizure variability, which, combined with their stable seizure rate across age, further supports the interpretation that their stress is more baseline or systemic rather than driven by a particular event. Taken together, these patterns call for more focused investigation into chronic stress and its neurological impacts on females, whose seizure activity appears less responsive to age or acute stress but is likely more tightly linked to sustained stress states.

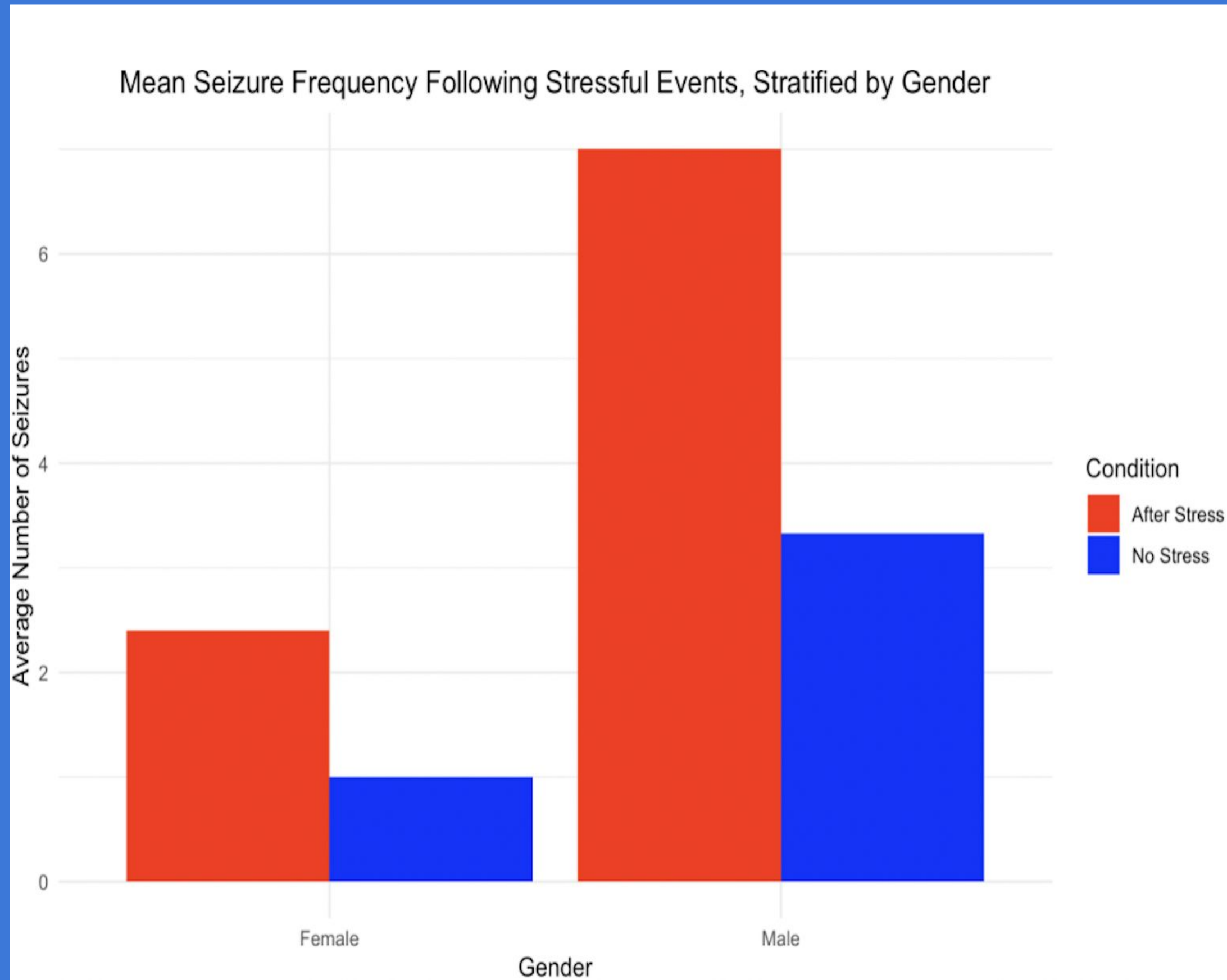


Figure 3: Males exhibit a sharper increase in seizure frequency following stress events compared to females, indicating a stronger acute stress response in men.

Source (Google Slides):

<https://docs.google.com/presentation/d/1d3WnQIRMIQzaeIKhGDjJnqwpguZawlo2vBBXuTBrT7U/edit>

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